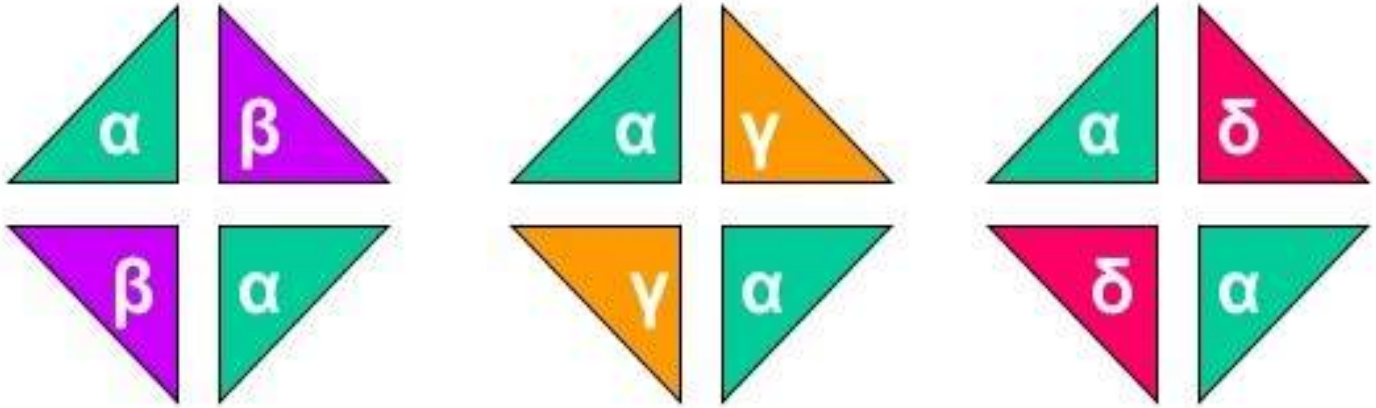
HEMOGLOBINOPATHIES

Dr. Humaira Aman Ali

Hemoglobins in normal adults



HbA HbF HbA_2

HEMOGLOBINOPATHIES

Definition:

" An inherited mutation of the globin genes leading to a qualitative or quantitative abnormality of globin synthesis".

OR

"Hemoglobinopathies are a group of genetic disorders caused by **production of a structurally abnormal hemoglobin molecule**; **synthesis of insufficient quantities of normal hemoglobin**; or, **rarely, both"**.

HEMOGLOBINOPATHIES

Qualitative hemoglobinopathy: production of hemoglobin with an altered amino acid sequence

- Sickle cell anemia (HbS)
- hemoglobin C disease (HbC)
- hemoglobin SC disease (HbS + HbC = HbSC)

Quantitative hemoglobinopathy: decreased production of normal hemoglobin

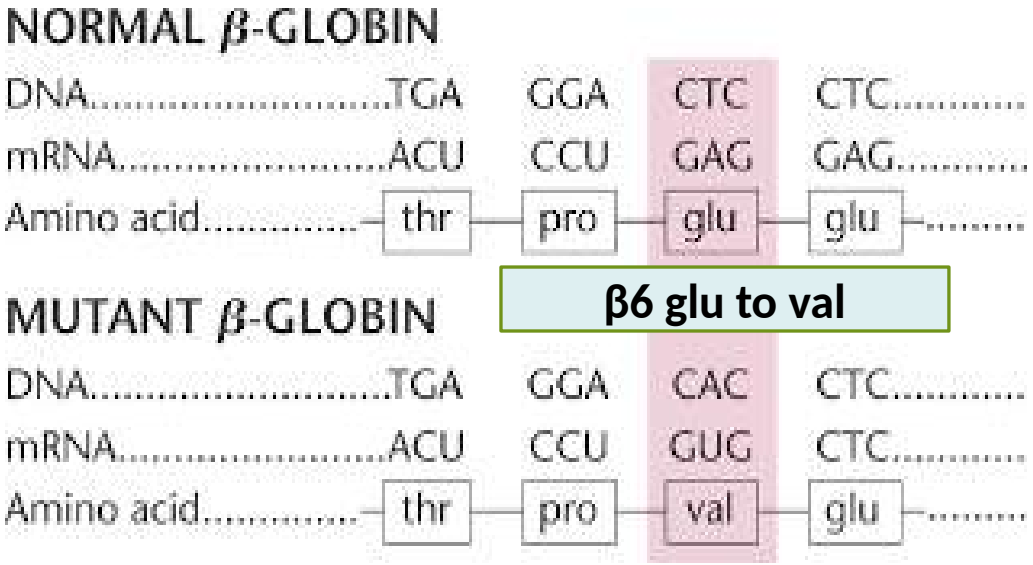
- Thalassemias

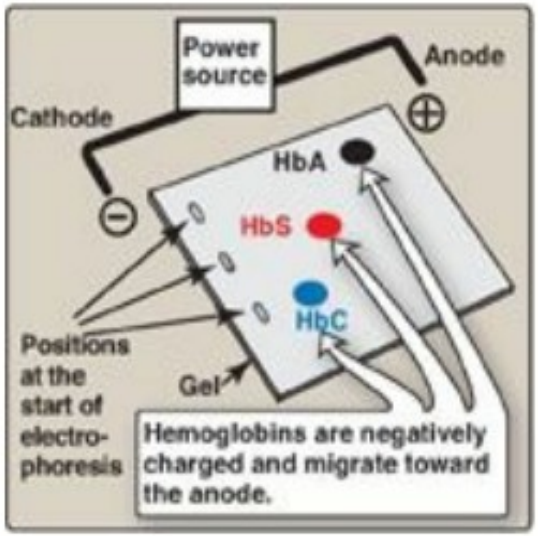
- **Abnormalities of hemoglobin structure cause hemolytic anemias that can have severe clinical consequences.**

HEMOGLOBIN S DISORDER (Sickle cell anemia)

General considerations

- An **autosomal recessive** genetic disorder, caused by a single nucleotide substitution in the gene for β-globin that code for synthesis of the β chains of the globin molecules.
- The mutant β-globin chain is designated βs, and the resulting hemoglobin, α2βS 2, is referred to as HbS.
- It occurs primarily in the African American population.





Sickle cell anemia

- Sickle cell anemia, the **most common** of the RBC sickling diseases.
- An infant does not begin showing symptoms of the disease until sufficient HbF has been replaced by HbS.
- Hemoglobin S polymerizes at low oxygen tension, forming **'Tactoids '** (a network of insoluble fibrous polymers) that distort the shape of red cells.
- Sickle shape red cells, repeated sickling episodes stiffen red cell membranes, making affected cells more subject to hemolysis.
- Small blood vessel occlusion and tissue infarction that results in "**Vaso-occlusive painful crises" (severe pain in limbs, back, chest and abdomen) often precipitated by infection or dehydration.**

- Repeated infarctions in the lungs and spleen; the spleen is characteristically congested and enlarged in childhood, but becomes progressively smaller through repeated infarcts and fibrosis (autosplenectomy).

- Severe chronic hemolytic anemia with associated hyperbilirubinemia.

- Increased susceptibility to infections, usually beginning in infancy.

- Other symptoms include acute chest syndrome, stroke, splenic and renal dysfunction, and bone changes due to marrow hyperplasia.

Normal and Sickled Red Blood Cells in Blood Vessels

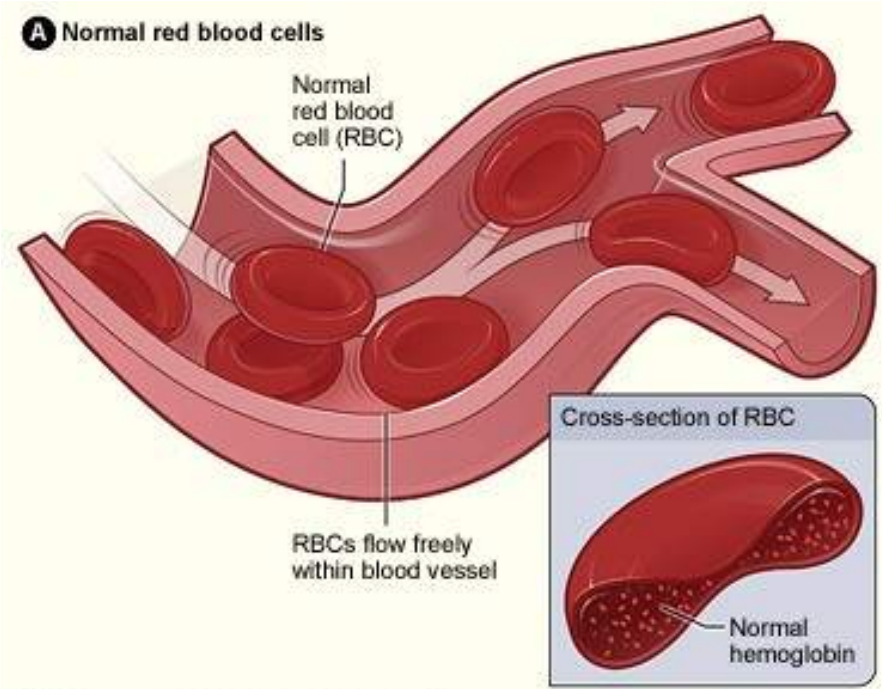
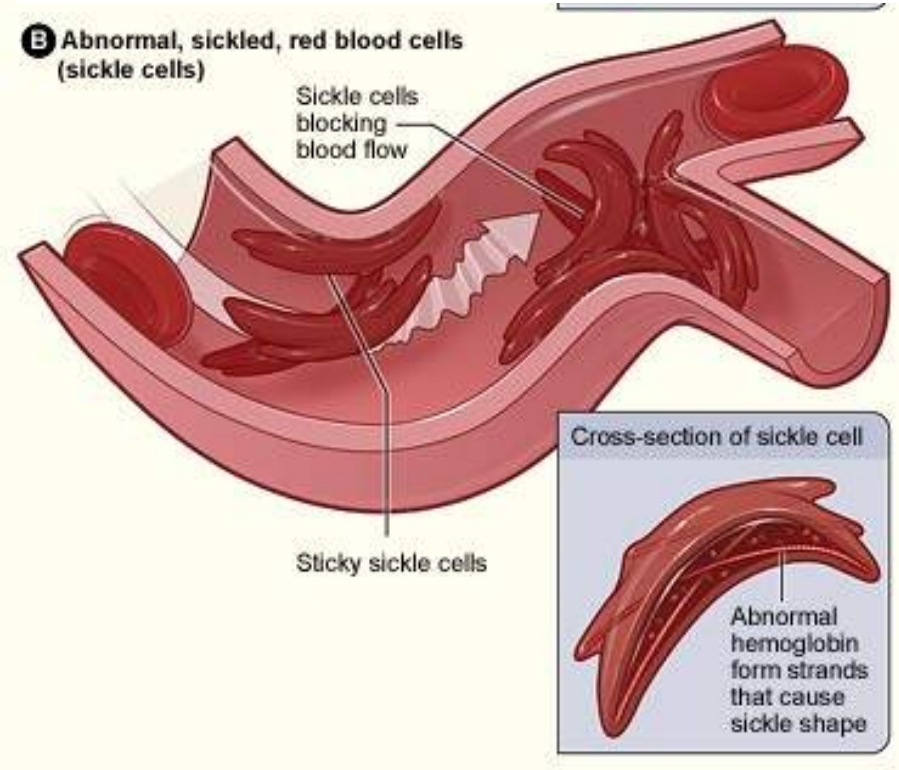


Figure A shows normal red blood cells flowing freely in a blood vessel. The inset image shows a cross-section of a normal red blood cell with normal hemoglobin.

Figure B shows abnormal, sickled red blood cells clumping and blocking the blood flow in a blood vessel. The inset image shows a cross-section of a sickled red blood cell with abnormal strands of hemoglobin.



Source from http://www.nhlbi.nih.gov/health/dci/Diseases/Sca/SCA_WhatIs.html

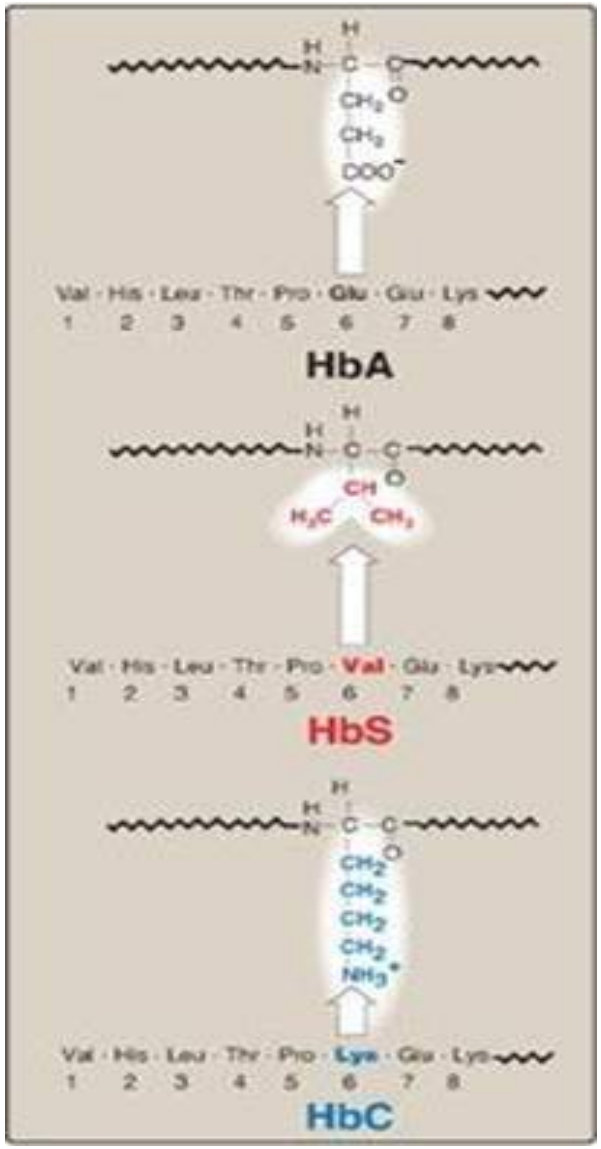
Sickle cell trait

- **Hemoglobin S in the heterozygous form (**have one normal and one sickle cell gene, the blood cells of such heterozygotes contain both HbS and HbA) **leads to sickle cell trait, which is generally without clinical consequence.**

- Heterozygotes for the sickle cell gene are less susceptible to the severe malaria caused by the parasite Plasmodium falciparum.

Hemoglobin C disease

- Like HbS, HbC is a hemoglobin variant that has a single amino acid substitution in the sixth position of the β-globin chain.
- A lysine is substituted for the glutamate
- Rare patients homozygous for HbC generally have a relatively mild, chronic hemolytic anemia.
- These patients do not suffer from infarctive crises, and no specific therapy is required.



Thalassemia

Definition:
a blood disorder passed down through families (inherited) in which the body makes an abnormal form of hemoglobin, the protein in red blood cells that carries oxygen

Formal Names:
Mediterranean anemia; Cooley's anemia; Beta thalassemia; Alpha thalassemia

This disease is inhereited

Starts from birth, you can get tested for it.

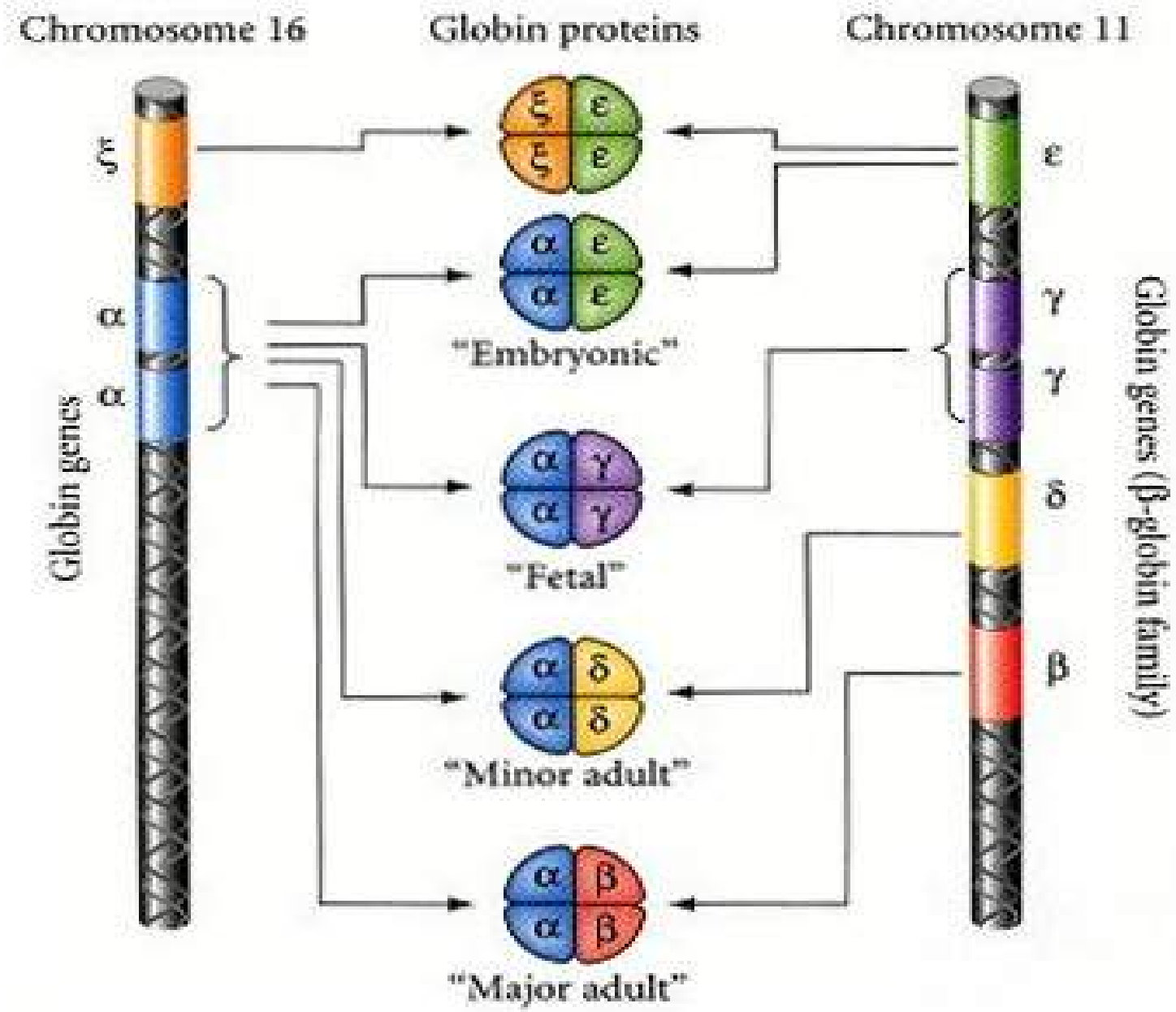
Affects the Bone Marrow

Treatment involves blood transfusions that must be given every 4 to 6 weeks to sustain life.

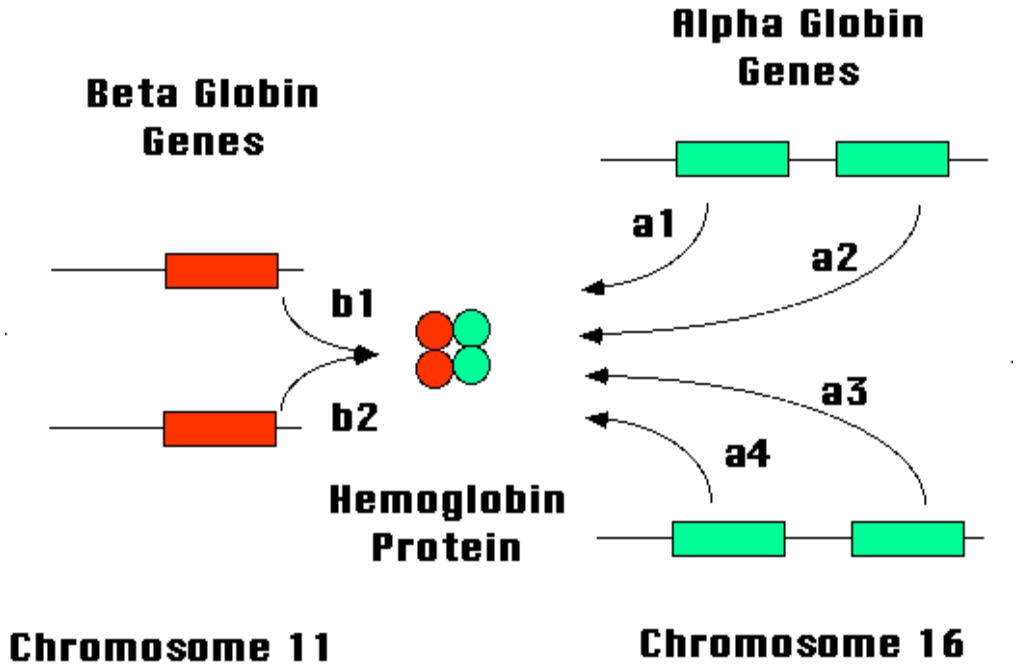
Can be acute or chronic, depends whether it's major or minor

THALASSEMIA (thal; α and β)

- This heterogeneous group of genetic disorders is characterized by deficient production of either α- or β-globin chains of hemoglobin.
- **Heme synthesis is unaffected.**
- Epidemiology:
  - Autosomal recessive disorder.
  - They are widespread throughout the world, occurring with high frequency in:
    - α- Thal is common in southeast Asia and in African Americans.
    - B- Thal is common in Mediterranean areas and in the United states.


Chromosome 16
Globin proteins
Chromosome 11
ζ
α
α
Globin genes
ε
γ
γ
δ
β
Globin genes (β-globin family)
"Embryonic"
"Fetal"
"Minor adult"
"Major adult"

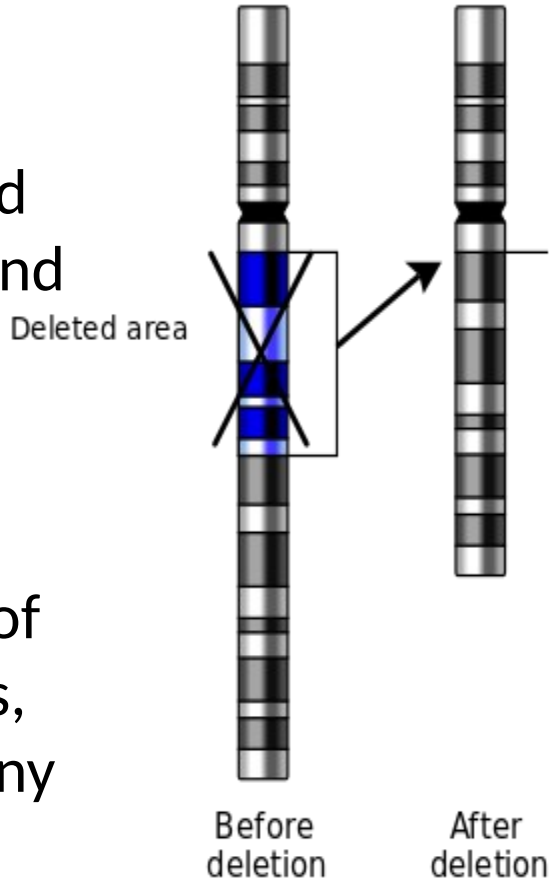
- The two chromosomes #11 have one beta globin gene each (for a total of two genes). The two chromsomes #16 have two alpha globin genes each (for a total of four genes).



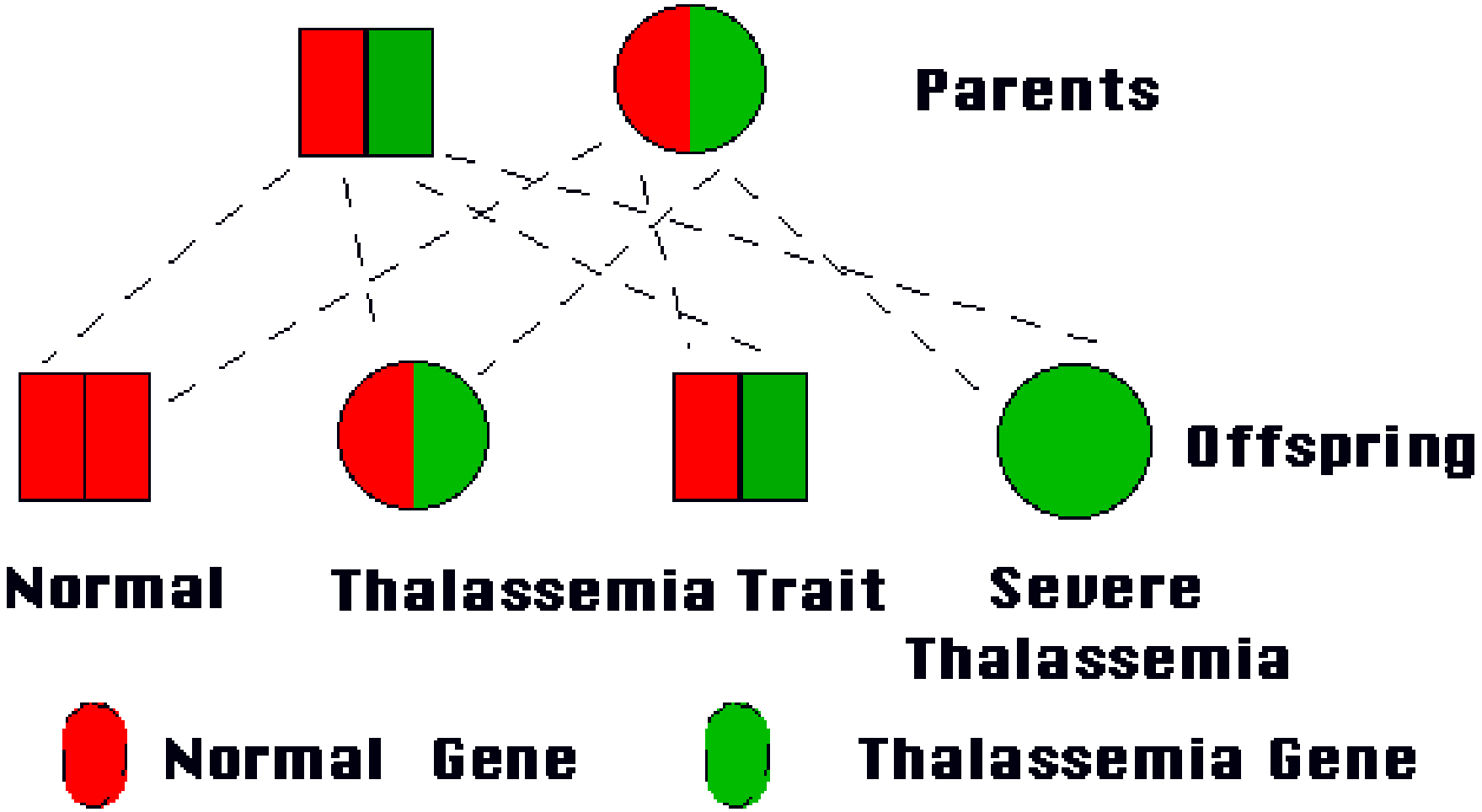
TYPES

- The **α-** and **β-talassemias** further classified into those producing no product (α_0, β_0) and those producing reduced amounts of product (α+, β+).

- A thalassemia can be caused by a variety of mutations, including entire gene deletions, or substitutions or deletions of one to many nucleotides in the DNA.



Thalassemia Inheritance
Two Carriers

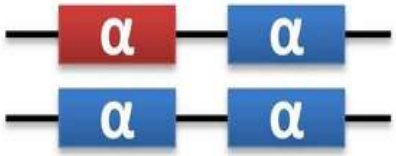


Alpha-thalassemia Genetics and Clinical Consequences

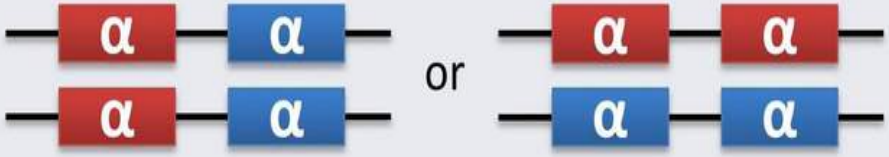
Normal



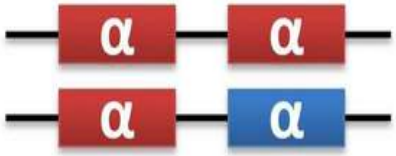
Carrier: Asymptomatic
No abnormalities



α-thal minor: Asymptomatic
Mild microcytic anemia



Hb H Disease: Symptomatic
Hemolytic and Microcytic anemia
Splenomegaly



Incompatible with Life
Hydrops Fetalis





Type of Hemoglobin

- Hb A $\alpha_2\beta_2$
- Hb A2 $\alpha_2\delta_2$
- Hb F $\alpha_2\gamma_2$
- Hb H β_4
- Hb Bart's γ_4

β-Thalassemias:

- **Decrease in β-globin chain synthesis**
  - **β^+ : reduced beta-globin chain synthesis,** β-thalassemia trait (β-thalassemia minor)
  - **β^0 : no beta-globin chain synthesis,** β-thalassemia major (Cooley's anemia)

- **More than 150 point mutations and several deletional mutations have been identified within and around the beta-globin chain gene.**

β – Thal Minor (beta^{+}-thalassaemia)

- Heterozygous inheritance
- Mild microcytic anemia
- Mild protective effect against falciparum malaria
- Decreased MCV, Hb, and Hct
- Normal serum ferritin
- On Hb electrophoresis
  - Decreased HbA
  - Increased HbA_2 and HbF

Beta-thalassemia minor with microcytes and target cells



β – Thal Major (beta^0-thalassaemia)

- **Severe hemolytic anemia**
  - **RBCs with α-chains inclusions are removed by macrophages in the spleen (marked spleenomegaly)**
  - **Ineffective erythropoiesis**
  - **Extramedullary hematopoiesis and distortion of skull, fascial bones and long bones because of erythroid marrow expansion**
  - **RBC morphology: marked microcytosis, hypochromic, target cells and extensive changes in size and shape**
  - **Increased HbF and HbA2**
  - **Generalized hemosiderosis.**


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Methemoglobinemias

- Oxidation of the heme iron in hemoglobin to the ferric (Fe3+) state forms methemoglobin, which cannot bind O2.
- Causes:
  - Certain drugs, such as nitrates
  - Endogenous products such as reactive oxygen species
  - Inherited defects
  - Deficiency of methemoglobin reductase, the enzyme responsible for the conversion of methemoglobin (Fe3+) to hemoglobin (Fe2+)
- The methemoglobinemias are characterized by “chocolate cyanosis” (a brownish blue coloration of the skin and mucous membranes and brown-colored blood) as a result of the dark-colored HbM.
- Symptoms are related to the degree of tissue hypoxia and include anxiety, headache, and dyspnea. In rare cases, coma and death can occur.


Thank you!